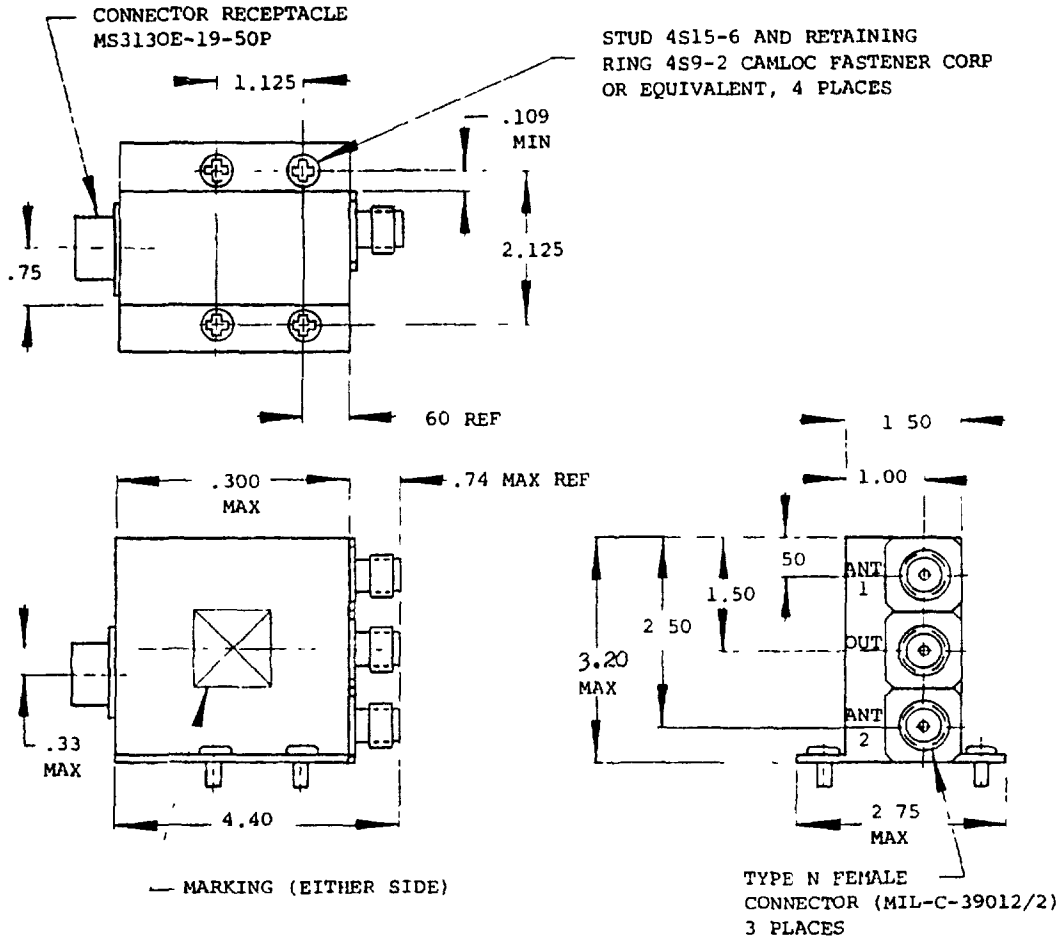
MILITARY SPECIFICATION SHEET

SWITCH, ANTENNA, RADIO FREQUENCY, SOLID-STATE

This specification is approved for use by the Engineering & Technical Support Division, HQ AFLC/LOIE, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of Specification MIL-S-83739.



8, 7	TO MICROSWITCH 13 AT2
1	+28 VDC IN
5	MIKE CONTROL
15, 13	GROUND
6	MAIN AGC IN
3	GUARD AGC IN (IF REQ)

8-GROUNDED	ANT 1 - OUT
8-OPEN	ANT 2 - OUT
8-7	SEARCH MODE

CONN WIRING

FIGURE 1. DIMENSIONS AND CONFIGURATION

FSC 5985

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INCHES	MM	INCHES	MM
.109	2.67	1.50	38.10
.33	3.38	2.125	53.98
.50	12.70	2.50	63.50
.74	18.80	2.75	69.85
.75	19.05	3.00	76.20
1.125	28.58	4.40	111.76

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.03 (.76 mm) for two place decimals and ±.010 (.25 mm) for three place decimals
4 The upper and lower surface of this area is not to be painted.
5. The spring shall be strong enough to hold the bolt free when disengaged from mounting plate.

FIGURE 1. Dimensions and Configuration - Continued.

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REQUIREMENTS:

Dimensions and Configuration See figure. 1
Weight· 1 pound max.
Finish· Black paint.
Mounting· Mounting provisions and mounting holes shall be provided that will interface with mounting plate MT-1995/A
Configuration. Single pole double throw (1P2T)
Remote Control: Provision shall be provided to remotely select either RF output on a full time basis by grounding pins on the power connector.
Nominal Impedance· 50 ohms.
Frequency Range· 225 to 400 MHz
Ant. Sample Rate· 1 0 ± 2 sec
Ant. Sample Time· 45 ± 5 ms.
Transmit Key:
If Hi (3V - 28V) normal receive operation
If Lo (0V - 3V) switch stops sampling and is locked in best previous receive antenna position
Primary Input Power 3 watts max, with the input voltage 28 ± 5V.
Transition Time· Less than 10 microseconds
Voltage Standing Wave Ratio (VSWR)
Antenna #1 = 1 5 1 max
Antenna #2 = 1.3 1 max
Isolation (crosstalk) 25dB min between output
Insertion Loss
Fail Safe (Antenna #2) .4dB max
Antenna #1 .80dB max
RF Power Handling Capability 100 watt peak power into a mismatched load with a VSWR rating of 2 1
MTBF Design for 20,000 hours, with a guaranteed minimum of 10,000 hours
Temperatures·
Operating -55°C to +121°C
Shock -40°C to +121°C
Environment MIL-E-5400 Class 2

'PPLICATION DATA

RF Sensitivity (for ARC-164 Rec) +10 -- -101 dBm
RF Diff Sensitivity (for ARC-164) 3dB

OTE. The switch covered by this specification sheet is substitutable for the following commercial types

Manufacturer's Code	Commercial Type
50501	CV-1002

Custodians
Air Force - 85

Preparing Activity
Air Force - 85

Review Activities
Air Force - 11, 99

(Project 5985-F465)

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-83739/2 (USAF) MIL-S-83739/2 (USAF) Switch, Antenna

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER FSC 5985

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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